3D Designer for ERPNext

Scope Document

Parametric 3D furniture design inside Frappe Desk, with automatic Manufacturing BOM, cutting plans, drawings, labels and CNC export.

Item	Detail
App	frappe_3d_designer
Platform	Frappe / ERPNext v16 (>=16.0.0, <17.0.0)
Runtime	Python 3.14+, Node 24
Dependencies	trimesh (Python); three@0.160.0 (JS)
Document date	6 November 2026
Status	Version 2 branch (v2), in active development

1. Purpose and background

Furniture and joinery manufacturers design in one tool, then re-key parts, materials and hardware into ERP by hand. This is slow and error prone. 3D Designer closes that gap: every design is a **Design Project** inside ERPNext, and its parts are converted into a Manufacturing BOM automatically. The designer, the cutting plan and the purchase/production data stay in one system.

2. Objectives

- Let designers model furniture and cabinets parametrically in the browser, with no external CAD licence.
- Generate a consolidated, accurate BOM (sheets, edge banding, hardware, connectors, operation minutes) on every save.
- Produce shop-floor outputs from the same model: technical drawings, cutting diagrams, part labels and CNC files.
- Import models and cut lists from other tools (GLB, OBJ, CSV, XLSX, JSON).
- Keep BOM revisions safe: submitted BOMs are versioned, never overwritten.

3. In scope

3.1 Design and modelling

Capability	Description
3D Designer page	/app/three-d-designer with 3D, Front, Top and Side orthographic views; drag to move with 5 mm snap; duplicate, rotate 90, delete.
Cabinet builder	Parametric dialog: size, plinth, shelves, doors, overlay or grooved back.
Furniture builder	Dining/coffee table, study desk, bench, chair, bed, bookshelf, TV unit with leg size/style, apron, seat height, slats and shelves.
Parts library	Design Library Component records (hinges, handles, legs, runners, profiles) inserted from the designer.
Part editing	Material, hardware item, shape (panel or cylinder), grain, size, qty, orientation, edge banding, exact position.
Visualisation	Section plane, explode slider, drilling overlay, presentation mode (PBR, wood grain, soft shadows), save render as PNG.

3.2 Workshop intelligence

- **Joint detection**: panels whose edge touches another panel's face become joints on save.
- **Connection Rules**: dowel, cam and dowel, screw, confirmat, shelf pin, groove, miter; thin-panel rule for back panels.
- **Automatic boreholes** on faces and edges, shown in 3D and exported to CNC.
- **Machining table**: manual Drill, Route, Pocket, Edge Profile, Miter and Groove operations per part.
- **Saw optimisation**: MaxRects nesting per sheet material with kerf, trim and grain direction.

3.3 ERP integration

Basis	BOM quantity per part
Sheet	Whole sheets from the cutting plan (stock UOM Nos)
Area	L x W / 1,000,000 x qty (m2)

Basis	BOM quantity per part
Volume	L x W x T / 1,000,000,000 x qty (m3)
Length	Longest side / 1000 x qty (m)
Nos	qty

- Wastage % added on top (except Sheet). Banded edges add the edge band item in metres. Joints add connector hardware and operation minutes.
- Same raw material across parts consolidates to one BOM row; hardware items are added per piece.
- Background job on save (hash-checked): creates the finished good item if missing, updates the draft BOM in place, or creates a new BOM version if the old one is submitted.
- BOM linked to Design Project (BOM.design_project custom field); live user notification; errors shown on the form.

3.4 Outputs

Output	Detail
Technical drawing	Front, side, top views, overall dimensions, part balloons, title block, Section A-A. Print format included.
Cutting plan	Sheet diagrams per material (print format and SVG).
Part labels	100 x 50 mm print format with QR code per piece.
CNC export (zip)	cutlist.csv, program.json, one DXF per part with layer-coded machining, sheet SVGs, drawing SVG.

3.5 Import

Format	Notes
GLB / GLTF	Embedded GLTF only. Each mesh is a part, named Part__Material. Size from oriented bounding box.
OBJ	Same naming rule.
CSV / XLSX	Cut list; English and German headers supported.
JSON	Saved from the 3D Designer.

3.6 Configuration masters

- **Design Material Map**: raw material item, basis, sheet size, kerf, trim, grain, wastage, colour, edge band item, aliases.
- **Design Connection Rule / Item**: connector spacing, hole sizes, hardware, operation minutes. Four starter rules installed.
- **Designer Settings**: company, item group, default rules, BOM default/submit options, model unit, optional operations.
- **Designer Operation**: Cutting, Edge Banding, Assembly etc. with workstation and minutes.

4. Out of scope

- Native PYTHA, SketchUp, Blender or Fusion project files (proprietary). Export GLB/OBJ or a parts list instead.

- Free-form or curved-part geometry: parts are rectangular panels or cylinders; rotation snaps to 90 degree steps for joints, drawings and CNC.
- Ray-traced rendering: presentation mode is real-time PBR.
- Free-form nesting of curved parts: nesting is rectangular only.
- Machine-native CNC formats (MPR, CIX, XXL, etc.): export is generic DXF/JSON; a postprocessor can be added on build_program().
- Pricing/quotation engine, sales order generation and production planning beyond the BOM (use standard ERPNext).

5. Data model

DocType	Role
Design Project	Main document: parts, joints, machining, sheet layouts, files, linked BOM
Design Part	Child: dimensions, material, grain, edges, position, hardware
Design Joint	Child: detected joint and its connector rule
Design Machining	Child: drill/route/pocket/profile/miter/groove rows
Design Sheet Layout	Child: nesting result per sheet
Design Material Map	Master: board/material mapping
Design Connection Rule / Item	Master: connector rules and hardware
Design Library Component	Master: reusable hardware and profiles
Designer Settings / Operation	Single settings and operation masters

6. Architecture

Module	Responsibility
api.py	get_design, save_design, generate_bom, get_materials
design_parser.py	JSON / CSV / XLSX / GLB / GLTF / OBJ to parts
bom_builder.py	Parts to consolidated BOM and operations; background sync
geometry.py / workshop.py	Part geometry, joints, connectors, boreholes, grooves
nesting.py	MaxRects saw optimisation
drawings.py / labels.py	Technical drawing, cutting plan SVG, QR labels
cnc_export.py	CSV / JSON / DXF export
public/js/designer_3d.bundle.js	Three.js editor and form preview

7. Users and roles

Role	Responsibilities
Administrator	Install, create items, material maps, rules, library, settings (one-time)
Designer	Create Design Projects, model or upload, review, save, produce outputs, revise
Production / Purchase	Consume the BOM, cutting plans, labels and CNC files

8. Workflow summary

- **Install**: get-app, build, install-app, migrate.
- **Setup**: Items, Material Map, Connection Rules, Library, Settings.
- **Create** Design Project; **get parts in** (design in browser or upload).
- **Edit and review** the model.
- **Save**: joints, holes, nesting, then BOM job.
- **Review BOM**; **produce outputs**; **revise** (new BOM version).

9. Assumptions and dependencies

- ERPNext v16 site with Manufacturing module enabled; raw material items exist and are mapped.
- Users have a modern browser with WebGL support.
- Uploaded GLB units match Designer Settings (usually metres).
- CNC operators map DXF layer names to operations once in their CAM software.

10. Acceptance criteria

- A cabinet or furniture piece designed in the browser saves and produces a BOM with correct sheet, edge band, hardware and operation rows.
- Editing parts and saving updates a draft BOM, or creates a new version when the BOM is submitted.
- Unmapped materials are reported clearly on the form instead of failing silently.
- Drawing, cutting plan, labels and CNC zip generate from the same model without manual re-entry.
- Imports from GLB, OBJ, CSV, XLSX and JSON create the expected parts.

11. Possible future work

- Machine-specific CNC postprocessors (MPR, CIX, XXL).
- Free-form nesting for curved parts.
- Arbitrary-angle rotation of parts.
- Quotation and costing generated directly from the design.
- Expanded furniture and hardware libraries.